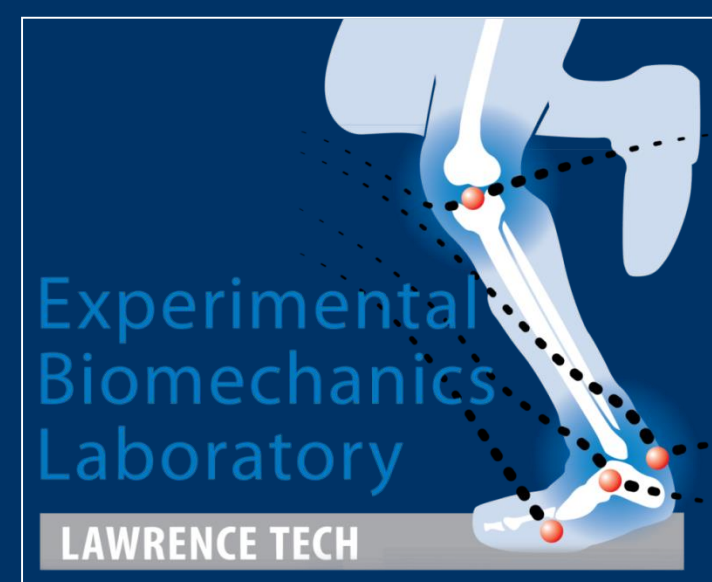
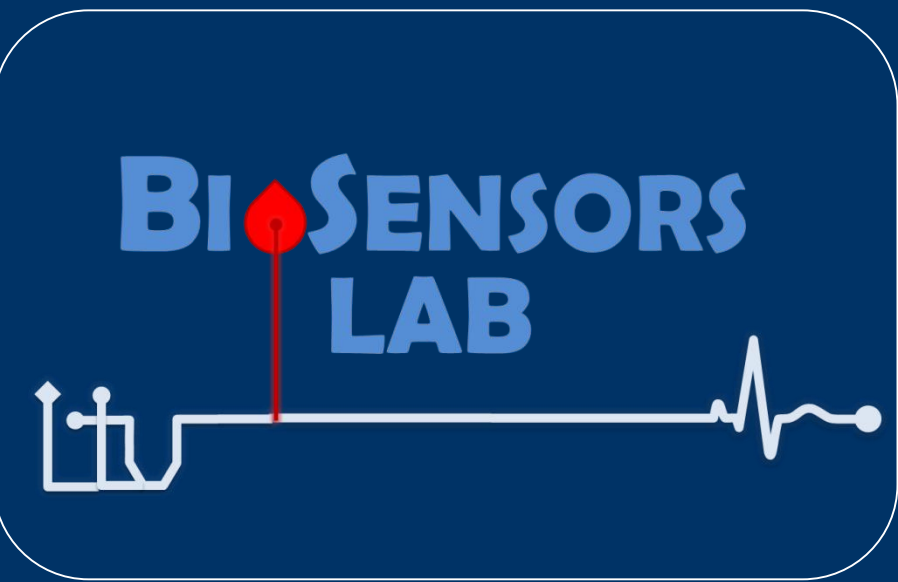


Fostering the Entrepreneurial Mindset through the Development of Multidisciplinary Engineering Learning Modules based on the “Quantified-Self” social movement: Pilot Modules Implementation

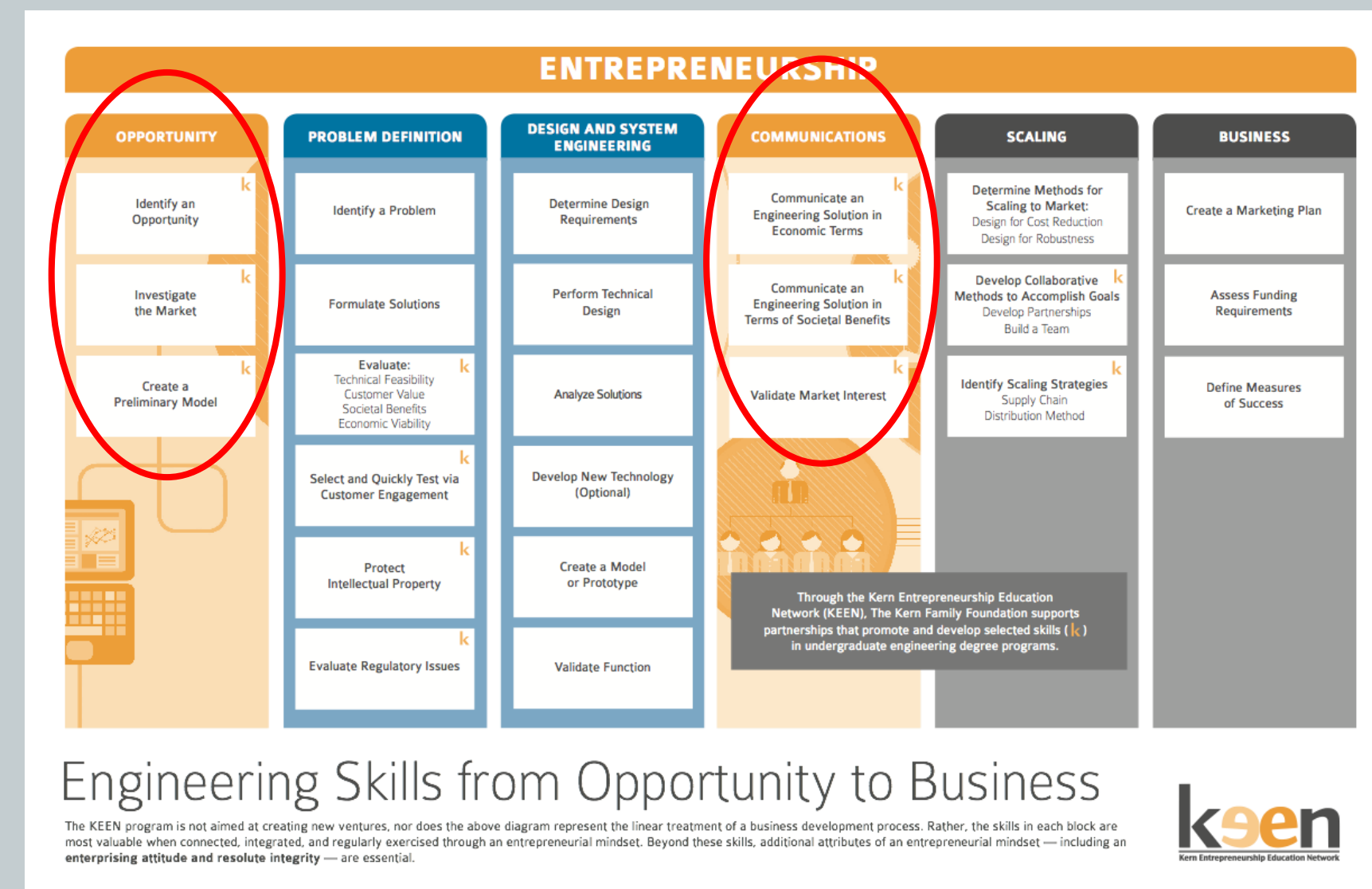
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Entrepreneurial Focus:

Traditional curricula and classes in biomedical engineering (BME) and other disciplines lack student learning experiences that promote entrepreneurship skills related to “Opportunity” and “Communications” (Fig. 1).



Engineering Skills from Opportunity to Business

Figure 1: KEEN Entrepreneurship Skills

Quantified Self:

The “Quantified Self”¹ (QS) social movement is a real world, exciting trend that can be used to motivate class modules that focus on entrepreneurial skills. QS is the convergence of miniaturization and biomedical testing technology, social media, and interest in personal improvement (Fig. 2).

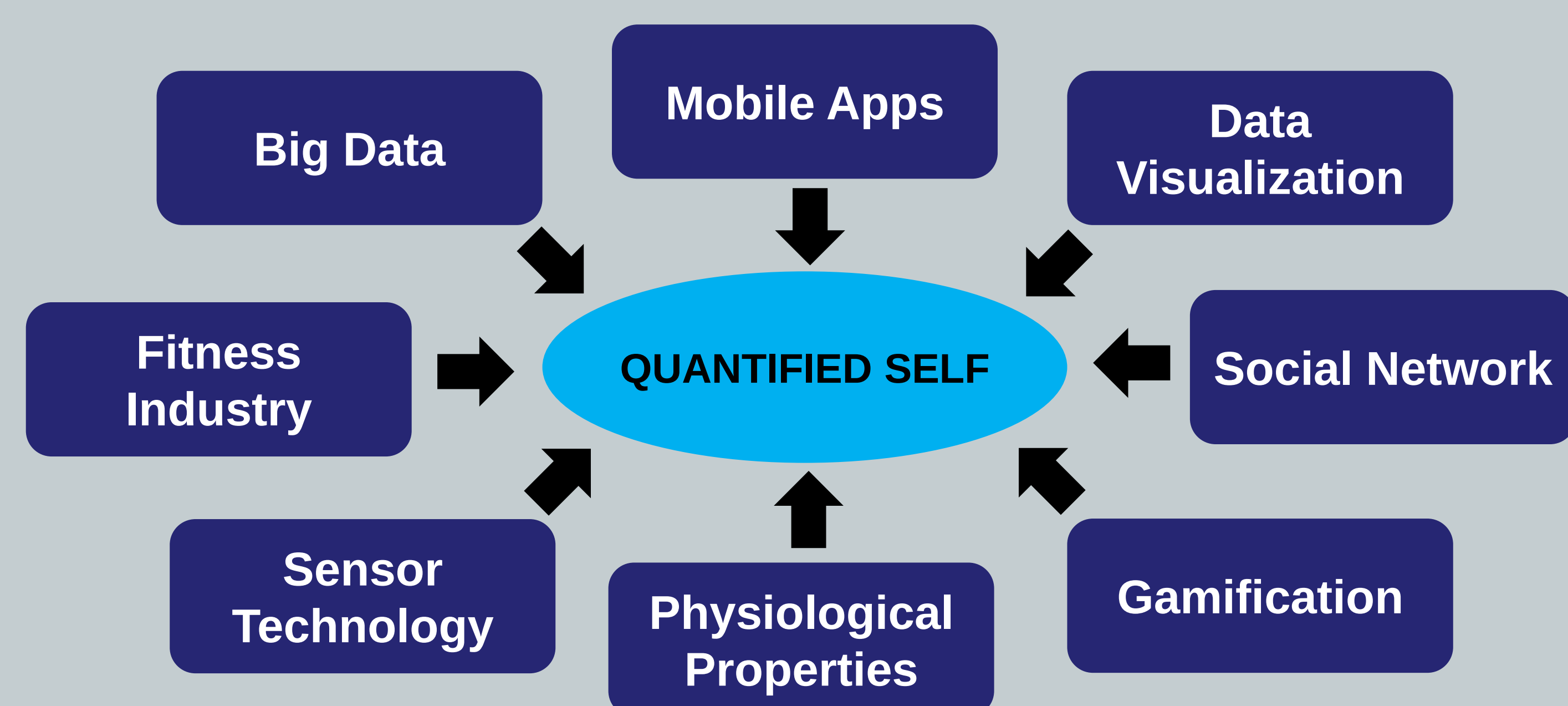


Figure 2: Graphical representation of the components of the QS movement

Project Goal and Target Curriculum

The project goal was to modify courses across the curriculum to promote entrepreneurship to train students to stop thinking only like an Engineer/Scientist and start thinking like a Product Developer. We targeted existing courses in BME and introduced modules that focused on entrepreneurial skills (Table 1).

Table 1. LTU BME Entrepreneurship Education

Level	Course	Status
Freshman	Fund. of Eng. Design Projects	Modified w/ LTU KEEN Grant
Freshman	Intro. to BME	QS Module Introduced
Sophomore	Multi-disciplinary, Project-based Eng. Studio	In Development w/ LTU KEEN Grant
Junior	BME Best Practices	QS Module Introduced
Junior	Medical Device Design	QS Module Introduced
Senior	BME Capstone Projects	Existing Coverage
Senior	Orthopedics	QS Module Introduced

Course modules were developed using Active, Cooperative Learning (ACL) and Problem Based Learning (PBL) pedagogical techniques using real world problems and current trends. Modules were implemented in four additional courses at the freshman, junior, and senior levels during the Spring 2014 semester. Direct and indirect assessment techniques were used to gauge the module’s effectiveness at changing students’ perceptions and improving their entrepreneurial capabilities.

Faculty Workshop (Summer 2014):

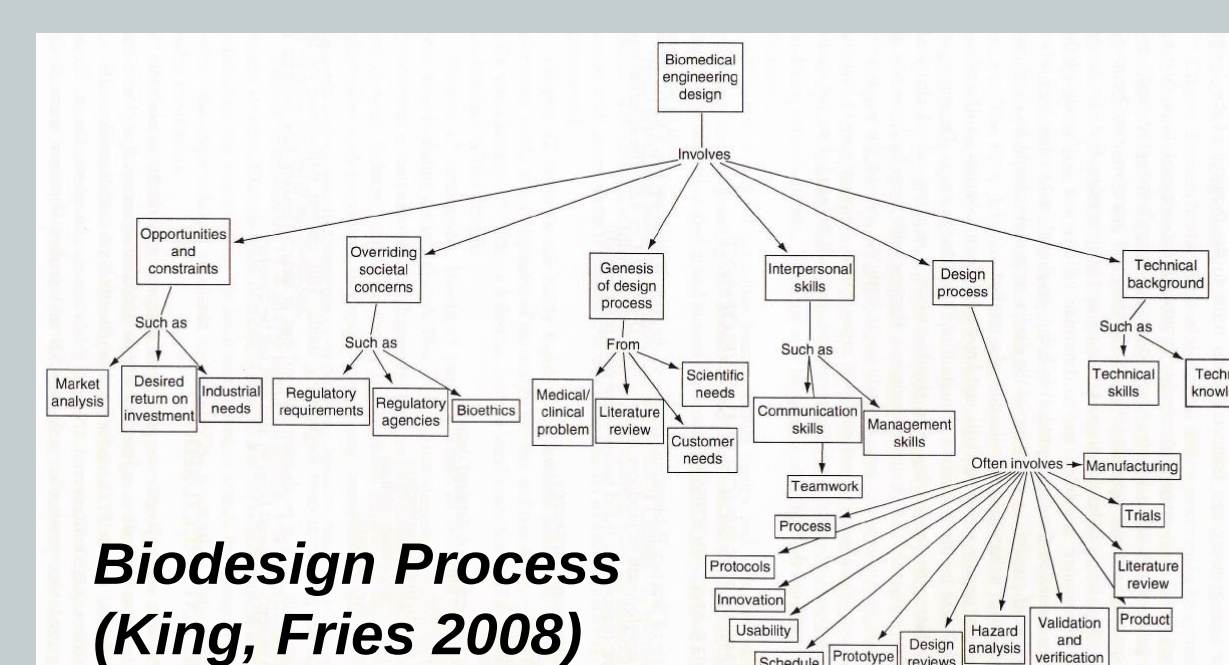
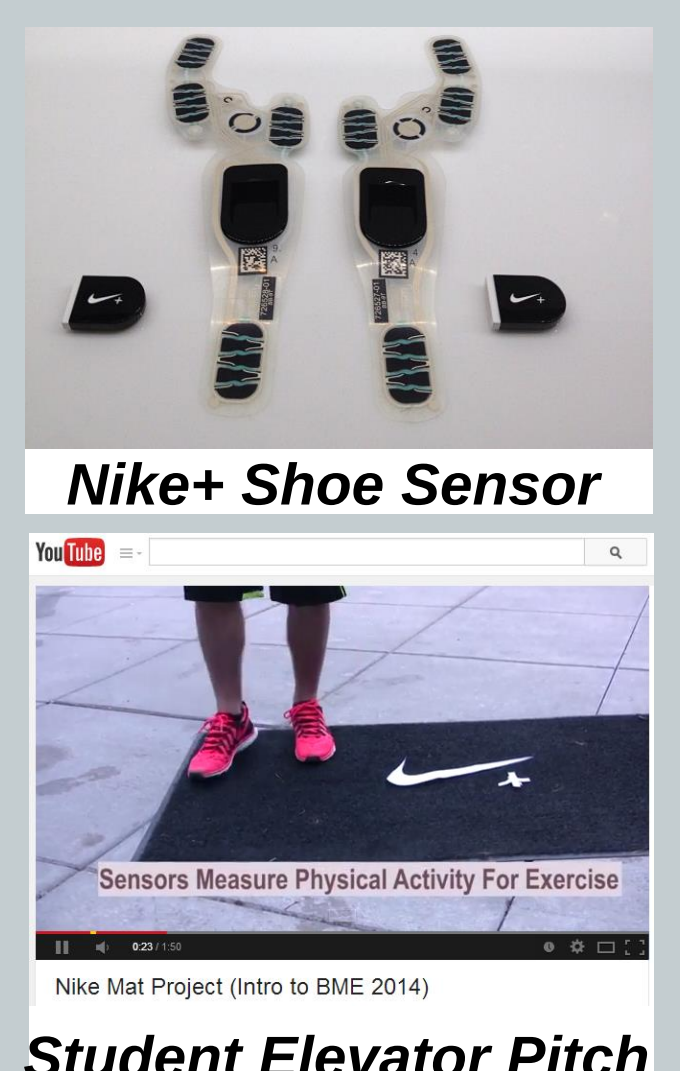
The coauthors are currently working with collaborators to develop modules for their specific courses using the examples, assessment results and student feedback from the pilot modules.

Pilot Modules:

Modules developed by the coauthors were multi-week assignments that were organized as PBLs combining several open-ended tasks, that built sequentially following previously completed work and the topics that were covered in class. They included smaller in-class ACL activities with individual or group homework assignments. At least two of the modules covered each of the entrepreneurship skills that were targeted (Table 2). Highlights of modules in each of the four course are described below:

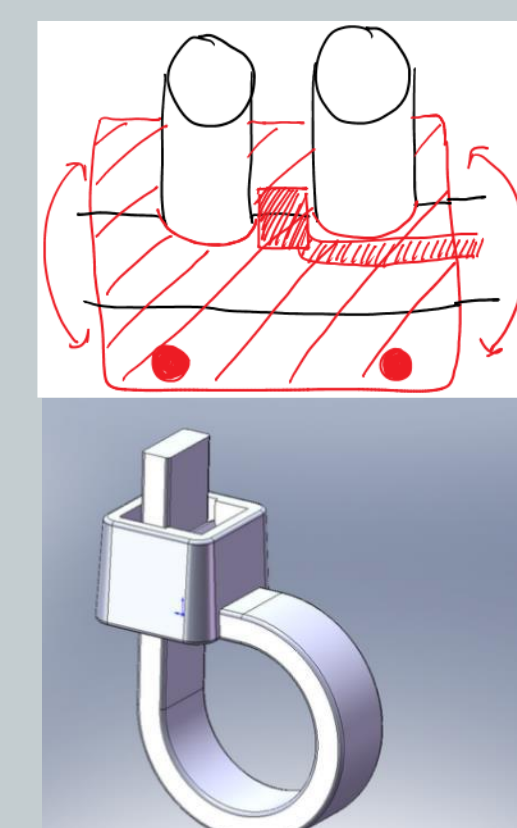
➤ Intro. to Biomedical Engineering (Meyer and Nasir):

- Opportunity Recognition through “Painstorming”
- Introduction to Quantified Self
- Using **Nike+ Sensor Suite** in a new application
 - Device Concept
 - Business Model and Market Potential
 - Elevator Pitch focusing on economic and social benefits of the idea



➤ Orthopedics (Meyer):

- Individual research into a specific technical issue in orthopedic practice
- Identification of a remaining challenge and opportunity recognition
- Recently approved medical devices and patents related to this issue
- Submission of a brief quality manual for **Good Manufacturing Practice**.



Design Prototypes

➤ Medical Device Design (Nasir):

- Class Presentations on opportunity recognition techniques
- Background investigation and presentation of clinical, technical and market feasibility of the idea.
- Elevator Pitch to explain the customer and economic value
- Mock 510k application
- **Design Prototype / Model**

Table 2. Entrepreneurship Skills coverage in pilot modules.

	Opportunity Recognition	Market Investigation	Create a Preliminary Model	Communicate solutions in terms of economic benefits	Communicate solutions in terms of societal benefits	Examine technical feasibility, economic drivers, societal and individual needs	Intellectual Property Protection	Regulatory Issues	Collaborate in a team setting
Intro. to BME	x	x	x	x	x				x
BME Best Practices		x		x	x		x	x	x
Orthopedics		x	x			x	x	x	
Medical Device Design	x	x			x	x		x	

Faculty Survey Results

Seven faculty members from different disciplines were asked to fill a survey about entrepreneurship and quantified-self. Some of the findings of the survey are below:

- All faculty members (100%) agreed that there was not enough emphasis on entrepreneurship in the current programs and recognized the need and importance of entrepreneurial skills for innovation in their respective fields.
- Among the entrepreneurial traits, **Creativity** was ranked highest (85%) followed by **Opportunity Recognition** (57%).
- All respondents (100%) chose **PBL** as the best mechanism to promote entrepreneurship through modules in existing courses
- According to the survey, all but one of the courses already includes some innovative learning modules focusing on ACLs, PBLs and Entrepreneurship.
- While faculty members were aware of the QS devices, most did not have initial ideas on how these could be used as a tool in their classes to teach entrepreneurship.